SHCHYGOLEVA INDUSTRIAL PUMPS

Maintenance Manual

SIP 80-200 Close-Coupled Centrifugal Water Pump

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Read this manual before you do maintenance on the pump. Follow the approved site safety procedures and keep the manual available to qualified personnel.

EDITORIAL CONTROL Engineering validation is pending. The equipment and company are fictional portfolio examples. This document must not be used to maintain actual equipment or presented as evidence of EU conformity.

Document control

Document identification

Field	Entry
Document ID	SIP-MM-001
Product	SIP 80-200 close-coupled centrifugal water pump
Issue	1 — editorial reconstruction
Market	European Union only
Scope	Standalone maintenance manual; separate installation and operating manual
Approval status	NOT RELEASED — engineering, risk assessment, translations and regulatory sign-off pending
Manufacturer address	[TO BE PROVIDED BY LEGAL MANUFACTURER]
Technical approver	[TO BE COMPLETED]
Compliance approver	[TO BE COMPLETED]

How to use this edition

This is a re-typeset editorial master. All source values, model designations and component identifiers remain provisional until approved against controlled engineering records. Square-bracketed HOLD statements identify information that cannot safely be inferred from the source.

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1 Introduction

1.1 Purpose

This manual provides maintenance information for the SIP 80-200 centrifugal water pump. It includes safety information, component identification, the maintenance schedule and maintenance procedures.

1.2 Applicability

This manual applies to SIP 80-200 pumps with serial numbers 21000 and higher, subject to confirmation against the approved product configuration. The serial number is shown on the pump casing data plate.

1.3 Personnel

• Operator — trained to operate the pump.

• Maintenance technician — trained and authorised to perform mechanical maintenance on pumps.

• Authorised electrician — qualified and authorised to isolate, verify and reconnect electrical equipment.

Where more than one role participates, each procedural step identifies the responsible person. Work must also comply with the employer’s approved site procedures.

1.4 Safety messages

Table 1: Signal words

Signal word	Meaning
DANGER	Hazardous situation that will result in death or serious injury if not avoided.
WARNING	Hazardous situation that could result in death or serious injury if not avoided.
CAUTION	Hazardous situation that could result in minor or moderate injury if not avoided.
NOTICE	Potential damage to equipment or property; not a personal-injury signal word.
NOTE	Supplementary information that is not a safety instruction.

Each safety message should identify the hazard, consequence and avoidance action. This document follows plain-language technical-writing principles; no formal ASD-STE100 compliance claim is made.

2 Safety

2.1 General safety rules

• Read the applicable maintenance instructions before starting work.

• Follow site-specific energy isolation, electrical safety and permit-to-work procedures.

• Prevent unexpected startup; verify the safe condition before maintenance.

• Allow hot surfaces and liquids to cool to an approved safe temperature.

• Depressurise and drain the pump before opening pressure-retaining parts.

• Use an engineered lifting method and verified lifting points for heavy assemblies.

• Use task-appropriate protective equipment.

• Keep guards and covers installed during operation.

• Do not run the pump dry.

ENGINEERING HOLD The safe isolation, depressurisation, lifting and restart methods require equipment-specific approval. Generic statements in this edition are not a substitute for the approved site and manufacturer procedures.

Table 2: Safety symbols

Symbol or topic	Meaning
General warning	Potential personal-injury hazard.
Electrical hazard	Dangerous voltage may be present in the motor and terminal box.
Hot surface	Pump casing and connected piping may be hot.
Moving parts	Do not touch moving components during operation.
Disconnect before maintenance	Isolate hazardous energy before servicing.

Table 3: Personal protective equipment

PPE	Use
Safety footwear	Handling and mechanical maintenance.
Protective gloves	Appropriate handling tasks; select for temperature and chemical exposure.
Eye protection	Draining, grease application and cleaning operations.
Other PPE	As required by the site risk assessment and safety data sheets.

3 Description and parts

3.1 General

The SIP 80-200 is a close-coupled, end-suction centrifugal water pump. The impeller is installed on the motor shaft. The pump therefore has no separate coupling or bearing housing. Water enters through the suction nozzle and leaves through the discharge nozzle. A mechanical seal limits leakage at the shaft passage through the backplate.

The back pull-out design allows the motor, impeller and mechanical seal to be removed as an assembly while the casing remains connected to the piping, subject to the approved removal and lifting procedure.

3.2 Technical data

Table 4: Technical data — provisional

Item	Source value — engineering verification required
Type	Close-coupled, end-suction centrifugal pump
Rated flow	80 m³/h
Rated head	32 m
Speed	2950 rpm
Motor	15 kW; 400 V; 50 Hz; three-phase; IE3
Bearings	Drive end 6309 C3; non-drive end 6209 C3
Maximum water temperature	80 °C
Maximum working pressure	10 bar
Mechanical seal	35 mm; carbon/silicon carbide; EPDM
Complete pump mass	Approximately 160 kg
Motor assembly mass	Approximately 120 kg

TECHNICAL HOLD Do not use the listed values as approved service limits until they have been checked against the motor nameplate, pump drawing, seal specification and engineering-controlled data.

3.3 Illustrated parts list

Figure 1 identifies the pump components. Match each figure callout to the item number in Table 5.

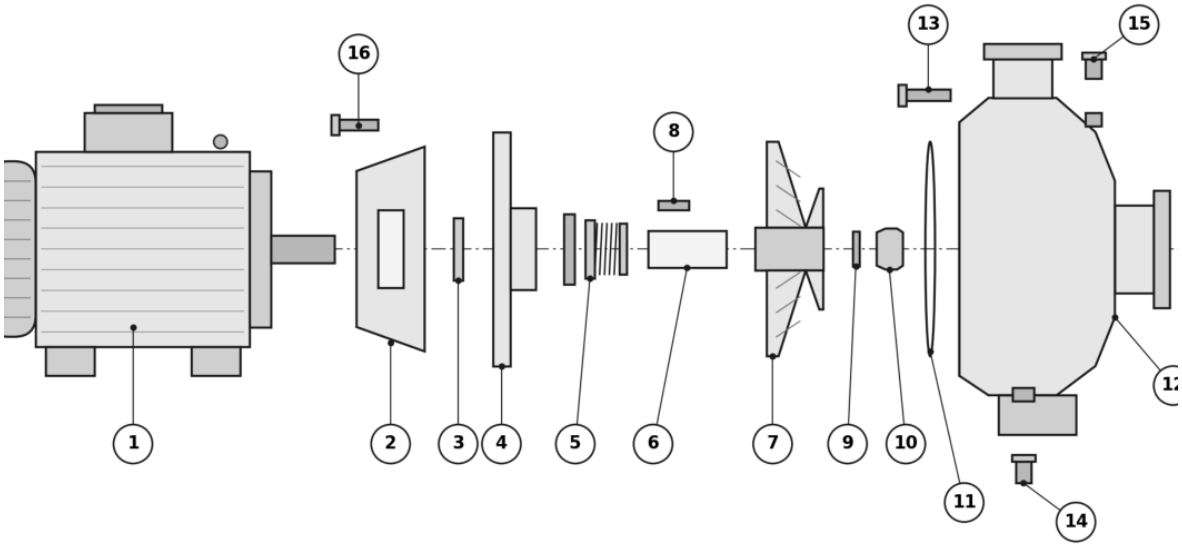


Figure 1: Exploded view of the SIP 80-200 pump

Table 5: Parts list

Item	Part number	Description	Qty	Spare
1	SIP-80-101	Electric motor, 15 kW, 2-pole, 400 V	1	
2	SIP-80-102	Motor adapter	1	
3	SIP-80-103	Water slinger	1	R
4	SIP-80-104	Backplate (seal housing)	1	
5	SIP-80-105	Mechanical seal, 35 mm, complete	1	R
6	SIP-80-106	Shaft sleeve	1	W
7	SIP-80-107	Impeller, 210 mm	1	W
8	SIP-80-108	Impeller key	1	
9	SIP-80-109	Impeller washer	1	
10	SIP-80-110	Impeller nut, M16, self-locking	1	R
11	SIP-80-111	Casing O-ring, EPDM	1	R
12	SIP-80-112	Pump casing	1	
13	SIP-80-113	Casing bolt, M12 × 40	8	
14	SIP-80-114	Drain plug, G 1/2, with seal	1	
15	SIP-80-115	Vent plug, G 1/4, with seal	1	
16	SIP-80-116	Adapter bolt, M10 × 30	4	

R = replace at each disassembly. W = inspect for wear and replace if required by approved criteria. Verify all identifiers and quantities against the approved bill of materials.

How to order spare parts

Specify the pump model, serial number, item number, approved part number and required quantity. Use manufacturer-approved replacement parts.

3.4 Mechanical seal parts

The mechanical seal comprises stationary and rotating components. Replace them as an approved complete set. Confirm the arrangement against the actual seal manufacturer’s documentation.

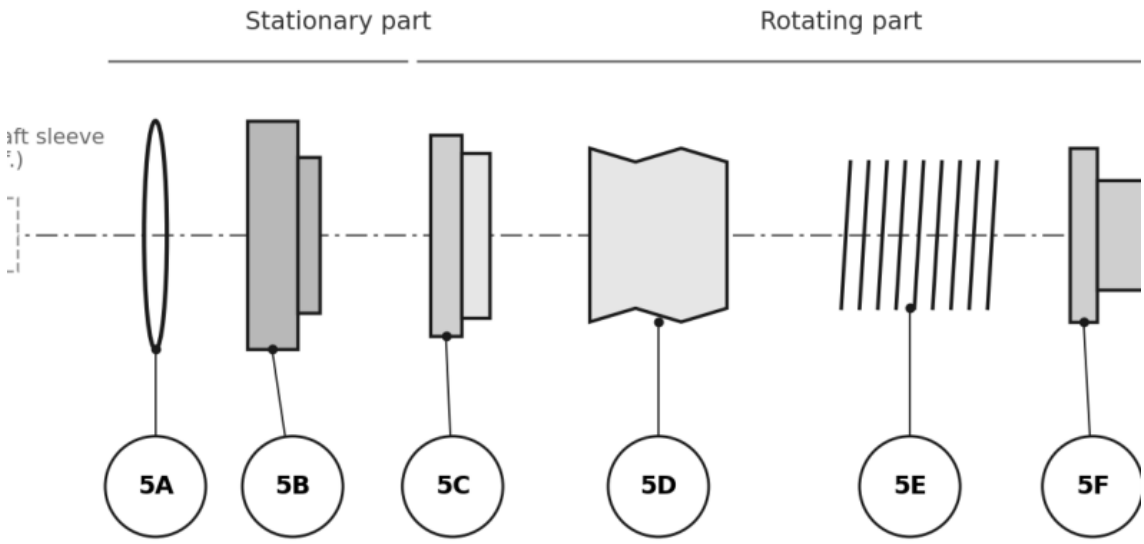


Figure 2: Exploded view of the mechanical seal (item 5)

Table 6: Mechanical seal parts

Item	Description	Function
5A	Seat O-ring	Seals the stationary seat in the backplate.
5B	Stationary seat (silicon carbide)	Stationary sealing face.
5C	Rotating face (carbon)	Rotating sealing face.
5D	Bellows (EPDM)	Seals rotating component to sleeve.
5E	Spring	Provides closing force.
5F	Spring retainer	Retains spring position.

4 Maintenance schedule

The source intervals below assume clean water at 80 °C or less and operation for no more than 16 hours daily. Intervals for continuous or severe duty must be approved by engineering. Record work and abnormalities in the equipment maintenance log.

Table 7: Maintenance schedule — intervals to be validated

Interval	Task	Reference	Role
Daily	Visual and operational inspection	5.2	Operator
Weekly	Clean motor fan cover and cooling fins	Site instruction	Operator
Monthly	Measure and record motor-bearing vibration	Site procedure	Technician
Monthly	Check pipe supports and casing fasteners	Approved instruction	Technician
2,000 operating hours	Lubricate motor bearings	5.3	Technician
12 months	Check motor insulation resistance	Approved electrical procedure	Electrician
12 months or seal leakage	Replace mechanical seal and casing O-ring	5.4	Technician
12 months	Inspect impeller and shaft sleeve	5.4	Technician

MAINTENANCE HOLD Intervals and acceptance limits must be confirmed against actual pump, motor and seal documentation. Record the date, running hours, findings, parts used, personnel and approval of return to service.

5 Maintenance procedures

5.1 Isolate and make the pump safe

Use the employer’s approved site-specific energy-control procedure before any work requiring isolation. The sequence below is an editorial outline; it must be replaced or approved by qualified personnel for the actual equipment.

Table 8: Personnel and equipment for procedure 5.1

Personnel	Equipment
Maintenance technician and authorised electrician	Approved lockout devices, tags, isolation verification equipment, valve restraints, drain container and appropriate PPE

DANGER Hazardous electrical energy can cause fatal injury. Only authorised persons may perform electrical isolation and verification. A start-button test does not prove absence of voltage.

1. Technician: Notify affected personnel and arrange a controlled shutdown.
2. Operator or authorised person: Stop the pump by the approved operating method.
3. Electrician: Isolate electrical energy using the identified disconnecting device and apply personal lockout controls.
4. Electrician: Verify an electrically safe condition using the approved absence-of-voltage test method before any electrical work.
5. Technician: Apply the required personal lockout and verify prevention of unexpected mechanical startup.
6. Technician: Isolate the suction and discharge lines using the approved valve and energy-control arrangements.

WARNING Hot or pressurised water can cause serious burns and eye injury. Confirm a safe temperature, verify zero pressure and control trapped liquid before opening the casing.

1. Technician: Allow the assembly to cool to the approved safe temperature.
2. Technician: Use the approved venting and draining method; collect and dispose of liquid safely.
3. Technician: Verify that all relevant stored energy has been released and document the safe condition.

ENGINEERING HOLD Approve actual disconnect locations, isolation verification, valve arrangements, drain/vent sequence, temperature limits and the required tools before releasing this procedure.

5.2 Daily inspection

Table 9: Personnel and equipment for procedure 5.2

Personnel	Equipment
Operator	Daily inspection record; appropriate PPE

WARNING Rotating parts and hot surfaces can cause injury. Keep guards installed and do not touch operating equipment.

1. Inspect the area beneath the backplate (4) for evidence of leakage. Report leakage and arrange inspection of the mechanical seal.
2. Listen for unusual pump or motor noise. If abnormal noise is detected, follow the approved shutdown and reporting procedure.
3. Read the discharge pressure gauge. Compare the reading with the engineering-approved operating envelope.
4. Check accessible casing joints, drain plug (14) and pipe connections for leakage without touching hazardous surfaces.
5. Confirm the motor cooling air path is unobstructed.
6. Record measurements, observations, abnormalities and action taken in the daily log.

TECHNICAL HOLD The source pressure band of 3.0–3.4 bar is not an approved universal acceptance criterion. Establish the correct pressure range for the actual installation and operating point.

5.3 Lubricate motor bearings

Table 10: Personnel, equipment and consumables for procedure 5.3

Personnel	Equipment	Consumables
Technician; authorised electrician as required for isolation	Calibrated grease gun, clean cloth, approved tools	Motor-manufacturer-approved grease and quantity

NOTICE Incorrect or excessive grease may damage bearings. Confirm grease compatibility, quantity, outlet-plug arrangement and whether lubrication is permitted while the motor is stopped or running.

1. Complete the applicable safe-isolation procedure 5.1 before touching components.
2. Clean accessible grease nipples and the surrounding area.
3. Check the motor manufacturer’s approved lubrication instructions and the bearing configuration.
4. Apply only the engineering-approved grease and quantity using the approved method.
5. Complete any required grease-outlet and run-in procedure in accordance with the motor instructions.
6. Return the pump to service using the complete approved procedure 5.5, not an isolated subset of steps.

7. Record lubricant identification, quantity, operating hours and results.

TECHNICAL HOLD Source values of 15 g (drive end), 10 g (non-drive end), NLGI 2 lithium-complex grease and 2,000 hours are provisional and require motor-manufacturer confirmation.

5.4 Replace the mechanical seal

This task includes inspection of the impeller, shaft sleeve and casing O-ring. The following sequence is a controlled editorial outline pending approval of the seal setting, lifting method and wear criteria.

Table 11: Personnel, equipment and consumables for procedure 5.4

Personnel	Equipment	Replacement parts
Two technicians; authorised electrician	Engineered lifting system, approved hand tools, calibrated torque wrench, cleaning materials	Complete seal (5), casing O-ring (11), self-locking impeller nut (10)

Table 12: Source torque values — unapproved

Item	Joint	Provisional torque
10	M16 impeller nut	90 N·m
13	M12 casing bolts	75 N·m
14	G 1/2 drain plug	30 N·m
16	M10 adapter bolts	45 N·m

WARNING The approximately 120 kg motor assembly presents a falling-load and crush hazard. Do not assume that a motor lifting eye is rated to lift the complete assembly. Use the engineering-approved lifting plan and verified lifting points.

Remove the motor assembly

1. Complete procedure 5.1 and verify the safe condition.
2. Electrician: Disconnect the motor connections using the approved electrical procedure.
3. Technicians: Attach approved lifting equipment at verified points and support the assembly.
4. Technicians: Remove casing bolts (13) using the approved method.
5. Technicians: Withdraw the motor assembly without imposing harmful loads on the shaft, seal or piping.
6. Technicians: Lower the assembly onto a stable, rated support.
7. Remove and discard casing O-ring (11).

Remove the impeller and mechanical seal

1. Secure the impeller using a method that does not damage components.
2. Remove and discard self-locking impeller nut (10); remove washer (9).

3. Remove impeller (7) using the approved puller and remove key (8).
4. Remove rotating seal components (5C–5F) and shaft sleeve (6).
5. Remove adapter bolts (16) and backplate (4).
6. Remove stationary seat (5B) and O-ring (5A). Discard the used seal components.

Inspect the removed components

1. Inspect shaft sleeve (6) for wear, grooves, scratches and damage against approved criteria.
2. Inspect impeller (7) for cracks, erosion and corrosion against approved limits.
3. Clean and inspect the backplate seal bore for damage.
4. Record all findings and replace components that fail approved acceptance criteria.

> ENGINEERING HOLD Provide measurable wear and rejection limits, inspection methods and approved component replacement criteria. Subjective descriptions such as “large erosion” are not sufficient.

Install the mechanical seal and impeller

> NOTICE Contamination or incorrect installation can damage the seal. Follow the seal supplier’s approved instructions for cleaning, elastomer lubrication, installation force, face handling and setting dimension.

1. Verify the new seal set and elastomers against the approved bill of materials.
2. Install the stationary seat and O-ring using the seal manufacturer’s approved method.
3. Clean sealing surfaces using approved materials.
4. Install the backplate and fasteners using verified tightening instructions.
5. Install the shaft sleeve and rotating seal components to the approved setting dimension.
6. Install the impeller key, impeller, washer and new self-locking nut.
7. Tighten approved fasteners to engineering-verified values using a calibrated tool.
8. Verify free rotation and all specified clearances and installation checks.

Install the motor assembly

> CAUTION Keep hands clear of closing flanges and pinch points when positioning the assembly.

1. Install a new casing O-ring (11) in the correct groove.
2. Use the approved lifting arrangement to align and insert the assembly.
3. Install and tighten casing bolts (13) in the approved sequence to the verified torque.
4. Remove lifting equipment only after the assembly is safely supported and secured.
5. Electrician: Restore electrical connections using the approved procedure.
6. Complete the full return-to-service procedure 5.5 and document the work.

5.5 Return the pump to service

Table 13: Personnel and equipment for procedure 5.5

Personnel	Equipment
Maintenance technician and authorised electrician	Approved tools, verified gauges, equipment inspection checklist and maintenance log

NOTICE Dry running may rapidly damage the mechanical seal. Do not start until the pump is correctly filled and vented.

1. Confirm that maintenance work is complete, components are secured and required fasteners are verified.
2. Install and secure the drain and vent fittings in accordance with approved instructions.
3. Verify the piping and valve arrangements for filling and priming.
4. Open the appropriate inlet path and fill and vent the pump using the approved method.
5. Set the discharge valve to the engineering-approved starting position.
6. Verify that all guards and covers are installed, tools are removed and personnel are clear.
7. Authorised persons: Remove their own locks and restore energy under the site lockout-release procedure.
8. Electrician: Restore electrical supply in accordance with the approved procedure.
9. Operator or authorised person: Notify affected personnel and start the pump.
10. Adjust operating valves using the approved startup procedure.
11. Check for leaks, abnormal noise, vibration, temperature and pressure.
12. Compare results with the approved operating limits; stop and report abnormal conditions.
13. Record the work, inspection results and authorisation to return to service.

ENGINEERING HOLD Confirm priming method, valve positions, lock removal responsibilities, acceptable leakage and pressure/vibration limits. Do not rely on the unverified source values or a partial restart cross-reference.

6 EU release and technical validation register

This register records matters that cannot be responsibly completed without product-specific source information. The manual is an editorial master, not a declaration of conformity or an approved maintenance procedure.

Release gates

Gate	Evidence required	Owner
EU legal framework	Applicable machinery legislation for placing-on-market date, other applicable acts and conformity assessment	Compliance
Manufacturer identification	Legal name, address, product/serial applicability and document approvals	Manufacturer
Product risk assessment	Maintenance hazard review and traceability of residual risks to instructions	Safety/engineering
Electrical isolation	Approved energy-control and absence-of-voltage verification method	Electrical engineer
Pressure and drainage	Approved temperature/pressure release and trapped-energy method	Mechanical engineer
Lifting	Assembly mass, centre of gravity, rated points and lifting plan	Mechanical engineer
Seal replacement	Supplier instructions, installation dimensions, material compatibility and wear limits	Seal supplier
Engineering data	Torque, grease, intervals, clearances and operating limits	Engineering
Language requirements	Destination Member State language plan and controlled translations	Compliance
Publication QA	Figure/table checks, accessibility, references, printed and digital preflight	Technical publications

Editorial sign-off

Technical writer: _____ Date: _____

Engineering reviewer: _____ Date: _____

Regulatory reviewer: _____ Date: _____

Publication approver: _____ Date: _____

Release condition: All technical and regulatory holds must be closed, and the final approved text must be checked against controlled source records before this document is issued for operational use.